Optimizing sales, analyzing the shopping trends and increasing Revenue of a small-scaled super market

Final Report for BDM Capstone Project

Submitted by

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1. Executive Summary and Title

Vaishnavi SuperMarket (VSM) is a small-scaled supermarket located in Malkajgiri district, Hyderabad. VSM is a B2C business, selling groceries and household necessities. The supermarket faced challenges in meeting sales targets and optimizing revenue. These issues could be addressed by adopting a data-driven approach to analyze sales trends, customer behavior and predicting pricing strategies.

The main aim is to increase sales and profitability by identifying purchasing trends and understanding customer behavior. Also, the impact of external factors like festival season. However, limited digital infrastructure posed a challenge. This report addresses the need for systematic sales analysis and optimized product pricing along with product strategies, to meet sales targets.

To overcome data limitations, transactions were manually recorded in excel. After creation of well-structures dataset with all the required columns, analysis of the data was carried out across multiple sheets in excel. Pivot tables and charts in excel were extensively used to analyze monthly revenue, expenses, and a product-wise and category-wise performances and findings were noted down. Further statistical and visual analysis were conducted in google colab using python, where ANOVA tests, monthly distribution plots and product category-wise comparisons were drawn. All of these gave a deeper understanding on the business sales, and had given significant patterns and trends which could help to satisfy the objectives of this project.

Key findings revealed top-performing categories like groceries and household essentials, high-margin items like frozen foods, and even a growing trend in UPI-based transactional payments. There were seasonal and festive impact on sales. For instance, March ice-cream sales had spiked up and in December and January, dry fruits sales had been maximum. Even though the non-festival periods had accounted for a high sales volume, the festival periods had significantly contributed to revenue. Also, December and January had seen a higher sales volumes compared to other months, emphasizing on the impact that the festive seasons create.

These have pointed out a need for targeted marketing, strategic promotions and bundling products and giving combo offers in order to only just increase sales, but to boost the profits. The findings of this study serve as a basis for strategic decision-making to achieve the business goals of Vaishnavi supermarket.

2. Detailed Explanation of Analysis Process/Method

a) Data Collection and Preprocessing:

The sales data was collected from the supermarket on a regular basis and the final data collected is from 01/12/2024 to 31/03/2025. All the objectives and methodologies were discussed with the owner, and data was collected only with his consent.

Although the data was stored in a software system, there was no functionality to download it, since the cost of software was high and the export option required payment which would add-on to business's expenses. To solve the problem, pictures of transactions were taken regularly and manually converted into Excel format.

b) Data Transformation:

Firstly, all the pictures we segregated into month-wise folders in my hard-drive. Random 10 transactions of each day had been selected for the analysis purpose of this project. Most of the products were retained in the roughly written list of products, and categorization of each product was manually performed. After this product name-category list was ready, few repeating products of 10 categories were finally selected for the analysis purpose. Meanwhile, selling prices and purchase prices of each of these products were written in a table format.

Now, the main excel sheet preparation had begun with drafting of columns- Date, Invoice ID, Customer ID, Product Name, Product Category, Selling Price and Purchase Price (each per unit) and the qty sold were all noted. Four transactions have been chosen for each day in order to maintain the length of the dataset. Calculation for Profit Margins and Total Revenue were prepared by applying formulas to few existing columns. Categorical columns like customer type, payment mode and festival indicator, were manually done inside the excel.

There was a lot of cleansing and organization done in the above process in order to ensure consistency. The final excel sheet was thus ready, and could be used for analysis. Once the dataset was formed, descriptive statistics was applied to understand key statistic measures like mean, median, mode, standard deviation and skewness within the data.

c) **Analysis:**

Various analyses were performed on the dataset to derive insights on the business trend, using pivot tables and charts in excel. The following analysis were conducted-

1. Revenue contribution analysis using Pareto Chart: Identified top 20% product categories that were contributing 80% of the revenue, using the pareto principle.
2. Purchase Price and Selling Price Comparison: Analyzed the per unit prices values of each product which helped to determine products with higher markup and profitability.
3. Month-wise Analysis: analyzed revenue, quantity sold and expenses of the business, each of these separately across various months.
4. Profit Vs Revenue Analysis: compared revenue and profits on a monthly and category-wise basis.
5. Customer Behavioral Analysis: Studied various customer patterns based on customer type, payment mode, and other external factors like festival seasons. The most frequent purchasers were drawn out by considering the customer ids and invoice ids to get the transaction frequency per customer.
6. Shopping Trend Analysis: Performed basic analysis like- Product Preference analysis, monthly Ice-cream sales trends, Festival impact on specific product categories. Used slider-based month-wise and product-wise comparisons.
7. ANOVA Analysis: This was performed on numerical columns, and analyzed few statistical measures like p-values and F-values, in order to get the significance of various groups. Further, this analysis was carried out in colab notebook for various non-numeric columns as well.
8. Product Performance Analysis: Compared and contrasted Total revenue and Sum of qty sold (product-wise). This was done in order to check for any relation between the two quantities.
9. Customer type Analysis: Based on the type of customer- returning or new, an analysis was carried out both using pivot tables and colab so as to find meaningful customer behavior, including factors like festival indicator.
10. Product category Vs Payment mode: There was an analysis of payment modes like UPI, Cash and Card carried out to know specific use cases of particular method used across categories.

11. Month-wise Payment analysis: To analyze a timely trend in payments made (in total number of transactions), a month-wise count of invoices was recorded for each payment method.
12. Relationship between Selling price and quantity sold: Across each of the product category, the average selling price and the sum of quantities sold were taken into account to check possible relation between the two.

3. Results and Findings

After in-depth analysis, the key results and findings that could be brought about are:

Category-wise Revenue and Qty Sold

1. The household essentials category sold the highest quantity (752units), but groceries and staples generated highest revenue (₹69,035), indicating higher per-unit value.
2. Stationery and snacks together contributed only 2.7% of total revenue (₹13,550 out of ₹3,14,182), showing a low revenue impact, despite having 10% of total units sold.

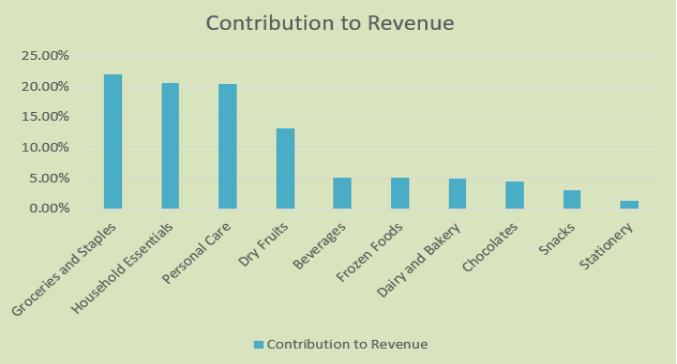

Fig 3.1 Contribution to Revenue

Purchase Price Vs Selling Price

1. Some products are sold below Purchase price, i.e they are sold at loss. For instance, Pista (₹275 → ₹270) and Heritage milk (₹30 → ₹29) are sold at loss, to attract customers and increase footfall. In such case, these need to be strategically sold with high margin items.

Category-wise Purchase and sales

1. Across all the categories, the total gross profit was ₹19,376.75, with groceries, household essentials and personal care categories accounting to 65% of it.
2. Stationery yields the lowest total revenue (₹1,855) and minimal gross margin (₹371).

Month-wise Revenue

1. January recorded the highest revenue (₹85,756), despite having only a slightly higher qty sold than march. This shows either high selling price or shift towards premium pricing.
2. February had the lowest sales volume (720 units) and the lowest total revenue (₹69,760).
3. March had bounced back with higher revenue (75541), almost touching January revenue.
4. December 2024, has seen the second highest revenue of all the four months with a revenue of ₹83,125 and 852 units sold.

Month	Sum of Quantity Sold	Sum of Total Revenue
Jan	873	85756
Feb	720	69760
Mar	854	75541
Dec	852	83125
Grand Total	**3299**	**314182**

Table 3.1 Monthly Total Revenue

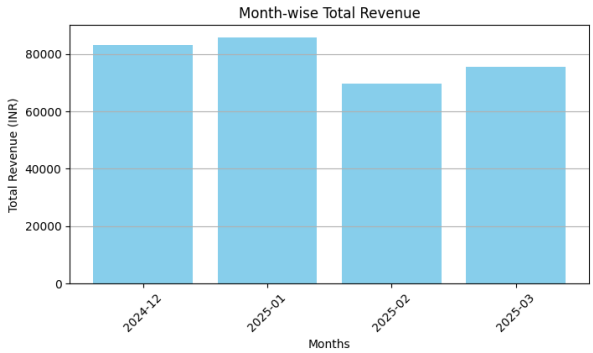

Fig 3.2 Month-wise Total Revenue

Monthly Expenses

1. Employee salaries account up to 53% of total monthly expenses, making it the largest expense the business is venting upon.
2. Miscellaneous expenses are just 13.9%, indicating minimal unpredictable spending.

3. Electricity expenses are higher, and together with misc. expenses, they contribute to one third of non-salaried expenses.

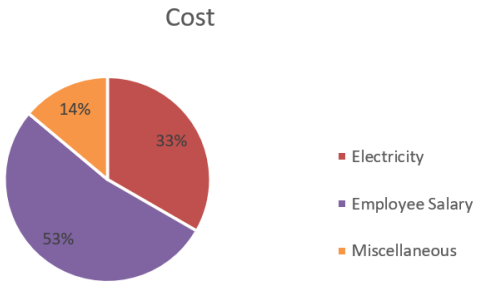

Fig 3.3 Monthly expenses

Profit Vs Revenue

1. Frozen foods and stationary items yield to highest average profit margins- up to 20.38% and 20% resp., indication top items in terms of profitability.
2. Dry fruits sales have reduced drastically across the months.

Customer Behavior

1. UPI dominated most of the transactions, accounting to almost 52% of all transactions (862 out of 1662).
2. Cash transactions decreased in February month (81), but recovered in March month to 116.
3. There was a steady decline in the Card transactions from December,224 to March,2025.

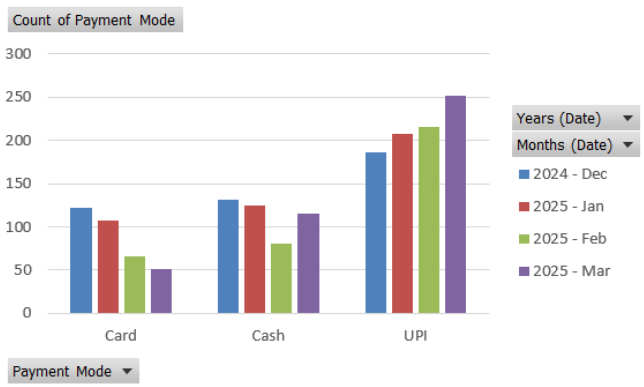

Fig 3.4 Month-wise Payment Mode analysis

Shopping Trends

1. Though Thumbs Up sold 22 units more than Coca Cola, the latter contributes 5% higher profit margin, suggesting stronger profitability despite having lower sales volume.
2. Out of total ice-creams sold, 42% (35units out of 83) sales were made in the month of March alone, showing a strong seasonal influence on frozen products. Kwality Walls had consistently higher sales in early months, but Cornetto overtook it significantly in March with 23 units vs. 12.

ANOVA

The purpose to perform ANOVA is to test whether there are any significant differences in the average values of a continuous variable (like revenue or quantity sold) across multiple categories (like product types, customer types, or payment modes). It helps you go beyond just visually interpreting charts and confirms whether observed differences are statistically meaningful or just random variation.

While Graphs are great for visual patterns like "Product A sells more than Product B." ANOVA is for validating insights statistically like "Is that difference in sales between products statistically significant?"

Below are few findings based on single-factor ANOVA test statistics:

1. Selling price per unit is on an average ₹95.33 as compared to than the average Purchase Price of ₹83.67. This indicates a modest pricing markup.
2. Purchase Price per Unit averages at ₹83.67, showing moderate variation in base costs across products (variance of 4185).
3. Selling Price per Unit averages ₹95.33 with a higher variance (5445), indicating pricing diversity, likely influenced by product type and demand.
4. Profit Margin shows low variance (0.0036) across products, suggesting consistent pricing strategies and a good cost control.
5. Quantity Sold has a low average (1.98) with low variance, indicating most items are bought in small quantities, with consistent behavior across products.
6. The average profit margin across all products is 11.7%, with a total revenue of ₹3.14 lakhs generated from 1662 transactions.

7. Total Revenue has a high average (₹189.04) and large variance (34393), indicating that some products contribute disproportionately to revenue.
8. The F critical value (2.37) is far less than the calculated F, confirming that observed variations aren't due to random chance.
9. There is a statistically significant difference between group means, with F (1152.72) much greater than F crit (2.37) and P-value < 0.05, confirming variability across product groups or months is meaningful.

Product Performance

1. Top 5 products alone (Pooja Oil, Nivea Deodorant, Dettol Liquid, Freedom Oil, and Kaju) contribute approximately 30% of the total revenue.
2. The product "Pooja Oil 1L" leads both in quantity (150 units) and revenue (₹33,000), contributing over 10.5% of total revenue, showcasing it as a high-performing staple.

Customer Type

1. Returning customers contribute a dominant 65.5% (₹205,948) of total revenue, significantly more than new customers (₹108,234 or 34.5%).
2. Festival season increased revenue across both segments, with festival-driven sales contributing ~27.4% (₹86,154) of overall revenue.

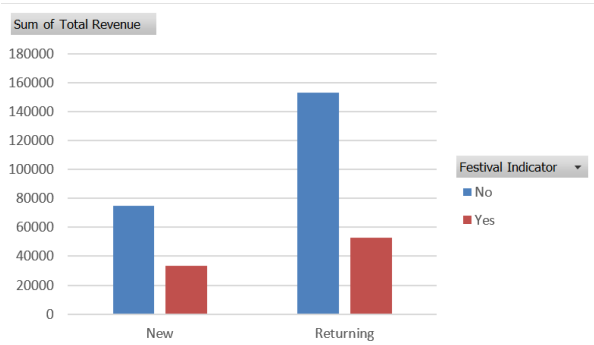

Fig 3.5 Revenue based on Customer Type, festival indicator

Product Category Vs Payment Mode

1. UPI is the leading payment method, used in ~52% of total invoices, while Card and Cash contribute ~20.8% and ~27.2%, respectively.
2. UPI dominates particularly in Household Essentials (192) and Groceries and Staples (187).

Month-wise Payments

1. Cash transactions increased moderately in 2025 (322) but still lag behind UPI, showing that while some still prefer cash, it's not growing at the same pace.
2. UPI usage surged significantly in 2025, growing from 186 in Dec 2024 to 676 in 2025, indicating a strong upward trend in digital adoption.

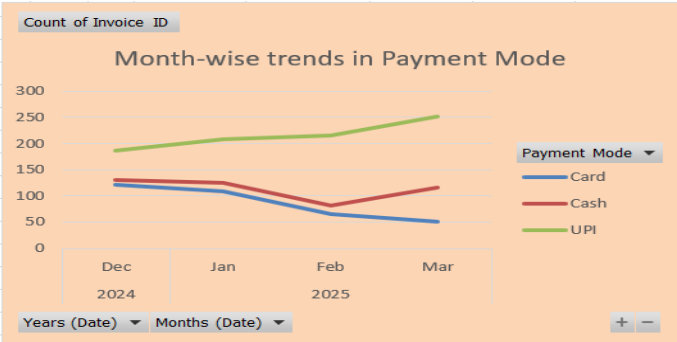

Fig 3.6 Month-wise Trends in Payment Mode

Relationship between Selling Price and Quantity Sold

1. High SP doesn't mean Higher sales. Categories like dry fruits with highest average selling prices have comparatively lower quantities sold indicating slow movement of premium-priced products.
2. Household Essentials (Qty: 752, Avg Price: ₹87.98) and Groceries and Staples (Qty: 701, Avg Price: ₹98.52) have high sales volume, indicating price-sensitive customer.

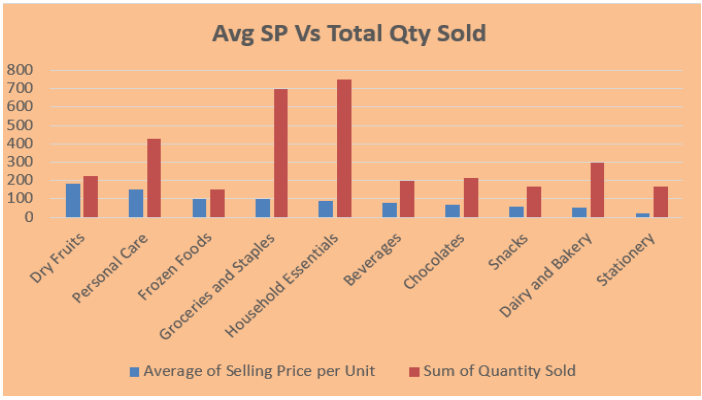

Fig 3.7 Scatter Plot of SP and Qty Sold

4. Interpretation of Results and Recommendations

Graphs and pivot tables have been drawn for every category and analyzed section-wise. The interpretations of the same has been noted below, along with any kind of recommendations that could be given to the business owner, so as to optimize sales and increase revenue of the small-scaled supermarket.

Category-wise Revenue and Qty Sold

1. Categories like Groceries have better revenue due to pricing or demand elasticity.
2. Low-revenue categories could be bundled with the above categories to boost profit, and improve the sales.

Purchase Price Vs Selling Price

1. A strategic way needs to be made in order to sell out low margin goods along with the high-margin ones.
2. The stationery items could be bundles up together to form one set that would contain a mixture of 2 or more stationery items.

Category-wise Purchase and sales

1. High difference in purchase and selling prices reflect a strong markup. Expanding SKUs in these categories will further develop on volume-based profits as well.
2. Reducing slow paced goods, and purchasing only when needed (on a seasonal basis) is highly recommended.

Month-wise Revenue

1. December and January months have seen very high revenues due to the influence of festivals. Trying to attract new customers during these months (customer acquisition) by providing offers and discounts, could further improve the customer footprints in upcoming months.
2. February has seen low sales due to lack of festive seasons, and fewer number of shopping days. To sustain customer engagement after the festive period, target marketing and customer retention schemes are quite necessary.

Monthly Expenses

1. Fixed costs are skewed towards employees' salaries, hence staff productivity needs to be analyzed and role optimization needs to be done in order to cut down on this part of the expenditure.

Profit Vs Revenue

1. High-margin categories like frozen foods and Stationery items strongly effect profits. Running targeted promotions like "Frozen delight Fridays" or "Smart Stationery Savings", could gear up the profits.
2. Increasing prices of combined-products by selling a mix of Dry Fruits together. Repositioning it as a health-focused bundle could easily justify the higher pricing, and could increase the margins.
3. Introducing combo offers and deals for snacks and beverages could increase overall sales volume, and thus improvising on revenue on these products.

Customer Behaviour

1. UPI payments made up to 51.9%, Card 20.8% and Cash 27.2% of total transactions, strongly indicating the customer shift towards digital payments especially the contactless and faster ones.
2. Introducing UPI based exclusive benefits like loyalty points.
3. Card usage has reduced steadily from 122 in December to 51 in march. But introducing Card-based schemes could attract some of the higher-bill categories.

Shopping Trends

1. Thumbs Up is locally preferred, but coca cola gives higher margins. Prioritize Coca Cola promotions over Thumbs Up to maximize overall beverage profitability.
2. Increase in March sales is likely due to warmer weather. Increase inventory and visibility of ice creams during peak months like March to capitalize on seasonal demand.
3. Cornetto showed a sudden gain in popularity compared to Kwality Walls. Launch targeted offers or combos for Cornetto variants would be very beneficial.

ANOVA

Single-Factor ANOVA (Analysis of Variance) is a statistical method used to compare the means of three or more groups to determine if at least one group mean is significantly different from the others. It helps identify whether observed differences in group averages are due to actual variation or just random chance.

In our project, we used single-factor ANOVA to analyze key sales variables across different product categories. It helped us validate whether variations in selling price, quantity sold, or revenue between categories are statistically significant, supporting better decision-making on pricing, product focus, and promotions.

1. Low variation in profit margin indicates pricing is uniform, but may not be optimized by category.
2. Implement category-specific pricing based on individual cost–profit analysis rather than flat markups.
3. Identify and prioritize these outperforming segments for promotional marketing and inventory stocking.
4. Tailor pricing, marketing, and stocking strategies based on the specific performance trends of each group.

SUMMARY				
Groups	*Count*	*Sum*	*Average*	*Variance*
Quantity Sold	1662	3299	1.984957882	0.973283532
Purchase Price per Unit	1662	139060.25	83.6704272	4184.970054
Selling Price per Unit	1662	158437	95.32912154	5444.995166
Total Revenue	1662	314182	189.0385078	34392.96059
Profit Margin	1662	194.4598228	0.117003503	0.003577388

ANOVA						
Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	40597950.58	4	10149487.64	1152.72466	0	2.373001786
Within Groups	73123702.33	8305	8804.780534			
Total	113721652.9	8309				

Table 4.1 ANOVA Test

Product Performance

1. High-selling, low-revenue products are likely daily essentials—important for customer retention and frequent footfall. Bundle such items (e.g., Rin Bar + Dettol) with mid-range products to increase basket value.

2. Premium pricing shows low volume, possibly due to price sensitivity. Introducing limited-time discounts could maintain a good perceivable value.

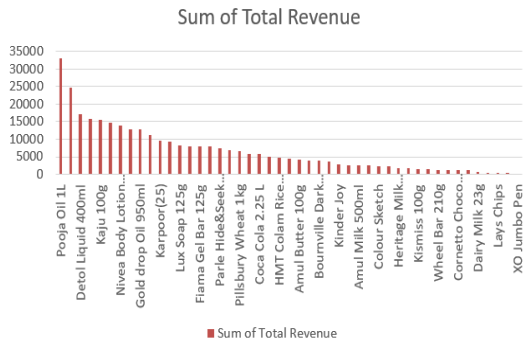

Fig 4.1 Product Performance – Revenue

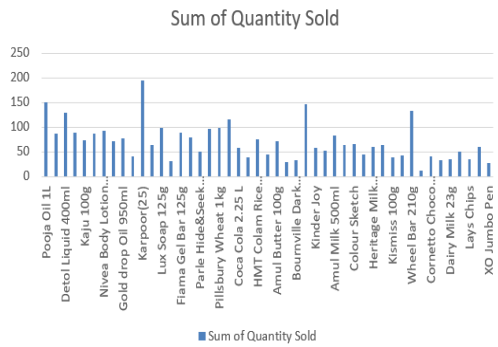

Fig 4.2 Product Performance – Qty Sold

Customer Type

1. Launch festival-specific onboarding offers to convert new customers into loyal returnees.
2. Introduce loyalty rewards or early-access festive deals to further capitalize on consistent spenders.

Product Category Vs Payment Mode

1. Low Card usage indicates possible lack of card incentives or POS limitations. Introduce card-specific combo offers or EMI options on higher-value products to increase adoption.

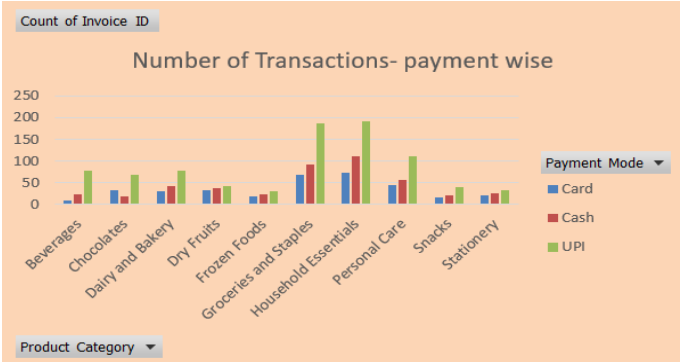

Fig 4.3 Payment-wise number of transactions

Month-wise Payments

1. Launch UPI-only flash deals which are limited-time offers or discounts that are exclusively available to customers who pay using UPI.
2. Launching QR-based in-store promotions. This is placing of QR codes in the store so that customers can scan using their phones to instantly access offers, discounts, or complete payments.

Relationship between Selling Price and Quantity Sold

1. Customers are price-sensitive, especially for frequently purchased items.
2. Consider offering smaller pack sizes for high-priced items to increase volume without lowering prices.
3. High-priced items have a niche demand, possibly bought by specific customer segments.
4. For popular and affordable categories, increase the inventory and make active promotions especially in peak months to capture the demand.

5. Explanation of Google Colab

An in-depth analysis was performed using python in google colab notebook. Here are the key points to observe and note from those graphs:

1. Beverages have seen gradual increment in months of Feb and March indicating seasonal demand. Dairy and Bakery have a constant demand throughout. Dry Fruits have seen a shoot up sales in Feb month indicating festival specific sales.

   Frozen Foods demand has been very less, unless festival season which got a very high revenue. Total Revenue of 'Personal Care' category has been constantly increasing, with highest revenue period in March, indicating festival impact (Holi).

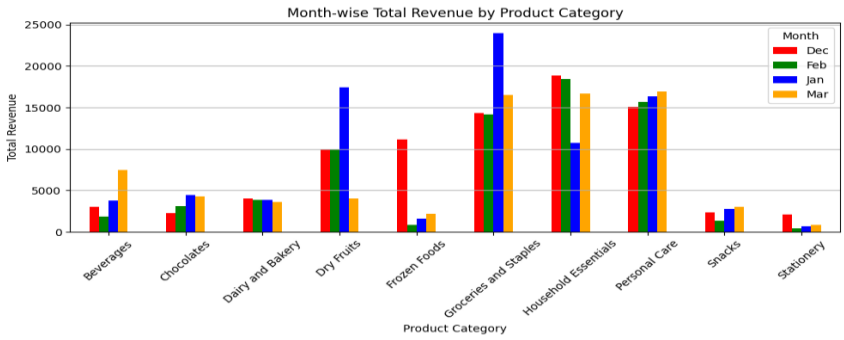

Fig 5.1 Month-wise Total Revenue

2. Top performer is Karpoor. It leads as the most sold item, nearing 200units, possibly due to high spiritual usage on daily basis. High demand for Daily essentials category. Items like Pooja Oil, Rin Bar, Wheel Bar and Dettol Liquid dominate the top sold items list, indicating a strong presence for household and hygiene product. Pillsbury wheat, Aashirvaad multigrain atta, and bambino vermicelli shows balanced sales, suggesting a consistent demand for staples and cooking items. Nivea body lotion and lux soap, though in top 10, but are at the lower end, possibly due to less frequent purchases or other brand options available to customers.

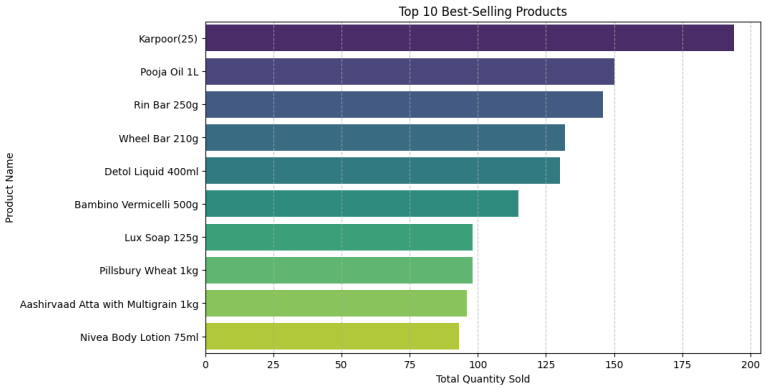

Fig 5.2 Top 10 Best-selling products

3. Revenue is not always propositional to quantity. Some categories like dry fruits and personal care show high revenue even at low quantity sold. This suggests a high-margin or premium-priced products.

   Most categories show peak revenue clustering around qty sold=3, indicating optimal purchase volume for many customers. Frozen Foods, Chocolates and Snacks show highest revenue almost at all quantity volumes especially when qty sold=4, indicating that these are revenue-driving items.

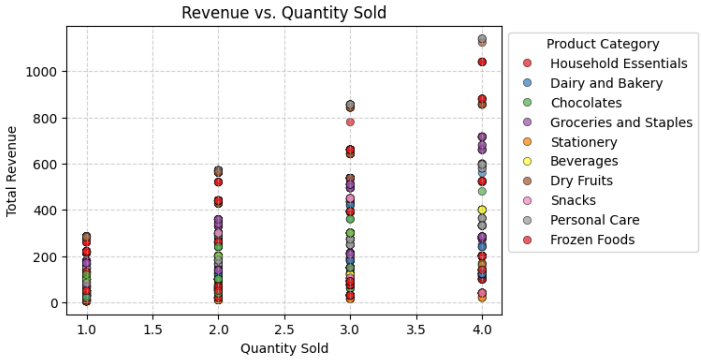

Fig 5.3 Revenue Vs Qty Sold

4. The median line of festival days is noticeably higher than that of non-festival days, indicating that average revenues are significantly better during festivals. The ANOVA test result shows an F-statistic of 8.07 and a p-value of 0.0046, which is well below 0.05. This confirms that the difference in revenue between festival and non-festival periods is statistically significant.
   Festival periods show a larger interquartile range and more extreme outliers, suggesting greater variability, possibly due to bulk buying. Since revenue spikes during festivals, the supermarket should consider targeted campaigns, festive discounts, or bundling offers during such times to maximize profits.

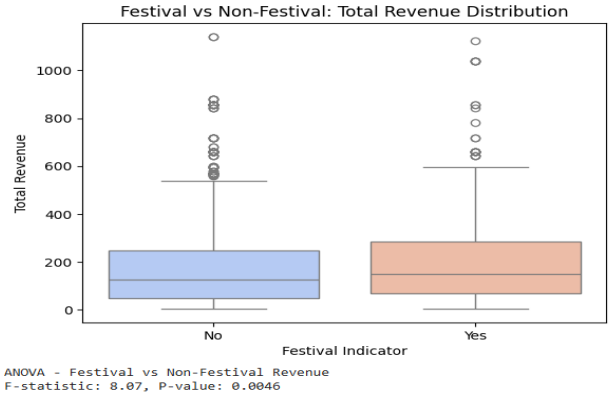

Fig 5.4 Festive Vs Non-festive Season

5. All 3 payment modes show a similar median revenue, indicating no strong preference in revenue generation based on payment method. Each type of method has a large number of outliers, indicating some transactions yield exceptionally high revenue, this is regardless of the payment mode. ANOVA results suggest that there is no statistical significance in revenue based on the payment mode. Card payments have slightly higher IQR and median, which might mean that there is a slightly higher margins, revenue made by card transactions.

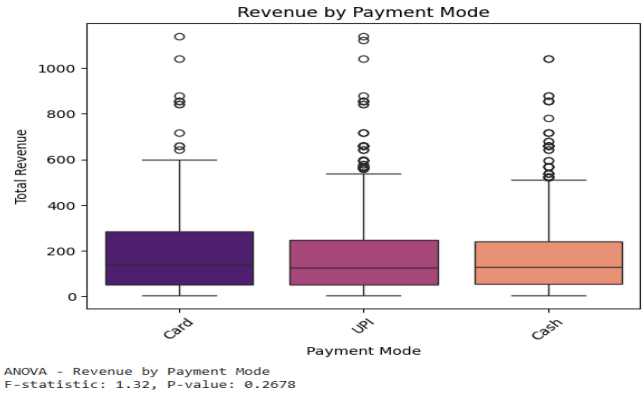

Fig 5.5 Payment-wise Revenue generated

6. Pricing Strategies

Below are few pricing strategies which I would like to suggest to the business owner. These strategies are designed to optimize pricing, attract a wider customer base, and boost sales volume. Implementing them can also help balance profit margins while remaining competitive in the market, thus focusing on growth and customer loyalty.

1. Bundling up Low Margin with High Margin Products can increase profits. For instance, 'Buy 2 Thumbs Up + 1 Chips at ₹X' would be a great way to cross-sell and balance margins.
2. Apply premium pricing to Dry Fruits and Frozen Foods during the festive season. Using scarcity marketing like 'limited stocks left over' to trigger urgency and justify the premium pricing.
3. Instead of cost-based pricing, assign prices based on the emotional or functional value to customers. Example: Position "Personal Care" items with wellness benefits at a slightly higher price, supported by health messaging.
4. Launching Combo Offers like 'Dairy with sweets', or 'beverages with snacks' during the festival season. We could consider Festival-themed packaging.
5. Demand-based pricing strategy, where price of a product is based on it's sales velocity. For instance, increase prices of fast-moving items like "Household Essentials" during peak demand weeks and offer discounts for slow-moving SKUs.
6. Offer a popular essential at a loss or breakeven (e.g., "Amul Milk 500ml" or "Tata Salt") to attract footfall, then upsell higher-margin complementary products like biscuits, snacks, or dry fruits in-store or on check-out screens. This is Loss Leader Strategy.
7. Introduce weekly or monthly subscription packs (e.g., milk + bread + eggs combo at a locked price). Builds customer retention, and gives predictable revenue and inventory flow.

8. Encourage digital payments by offering special discounts or cashback on UPI transactions (e.g., "₹100 off on orders above ₹2000 via UPI only"). This not only speeds up checkout but also reduces cash-handling effort. Tie this with loyalty rewards, frequent UPI users can unlock extra benefits like early access to festive deals or bonus points.
9. Using Tiered pricing: Economic, standard and premium, where USP could be 'buy more, save more'.
10. Introduce limited-time daily discounts during low footfall hours (e.g., 2–5 PM) called "VSM Happy Hours." Offer 5–10% off on selected essentials or fast-moving items. This helps to clear inventory quickly, balance customer flow, boost sales during slower periods and creating urgency with in-store announcements or SMS alerts like: "Flash Deal: 10% off on all dals between 2–5 PM today!".

7. Important Links

Here are few important links for originality and reference:

a) Datasheet Link: click here
b) Colab Link: click here
c) Gdrive: click here